

	Sentence	Class
1	All Fourier Transform input data is centralized on a single Backend processor via MC defined in BE	1
2	User interfaces comply with defined requirements and WCAG accessibility standards	0
	...	...
534	BE reports status, errors, and debug information to MC.	1
535	Navigation elements are self-descriptive and easy to understand.	0

Sentences from the dataset: **0 Non-functional requirement / 1 Functional requirement.**



Preprocessing



	Sentence	Preprocessed_sentence	Class
1	All Fourier Transform input data is centralized on a single Backend processor via MC defined in BE	fourier, transform, input, data, centralized, single, backend, processor, via, mc, defined, be	1
2	User interfaces comply with defined requirements and WCAG accessibility standards	user, interface, comply, defined, requirement, wcag, accesible, standard	0
	...	...	...
534	BE reports status, errors, and debug information to MC.	be, report, status, error, debug, information, mc	1
535	Navigation elements are self-descriptive and easy to understand.	navigate, element, self, descriptive, easy, understand	0

Sentences from the dataset: **0 Non-functional requirement / 1 Functional requirement.**

	Sentence	Preprocessed_sentence	Class
1	All Fourier Transform input data is centralized on a single Backend processor via MC defined in BE	fourier, transform, input, data, centralized, single, backend, processor, via, mc, defined, be	1
2	User interfaces comply with defined requirements and WCAG accessibility standards	user, interface, comply, defined, requirement, wcag, accesible, standard	0
	...	...	...
534	BE reports status, errors, and debug information to MC.	be, report, status, error, debug, information, mc	1
535	Navigation elements are self-descriptive and easy to understand.	navigate, element, self, descriptive, easy, understand	0

Sentences from the dataset: **0 Non-functional requirement / 1 Functional requirement.**



Get the vocabulary



Vocabulary

[fourier, transform, input, data, centralized, single, backend, processor, via, mc, user, interface, comply, defined, requirement, wcag, accesible, standard, ... , be, report, status, error, debug, information, navigate, element, self, descriptive, easy, understand]



Get the vocabulary using a Pareto-based frequency cutoff



Vocabulary using Pareto-based frequency cutoff

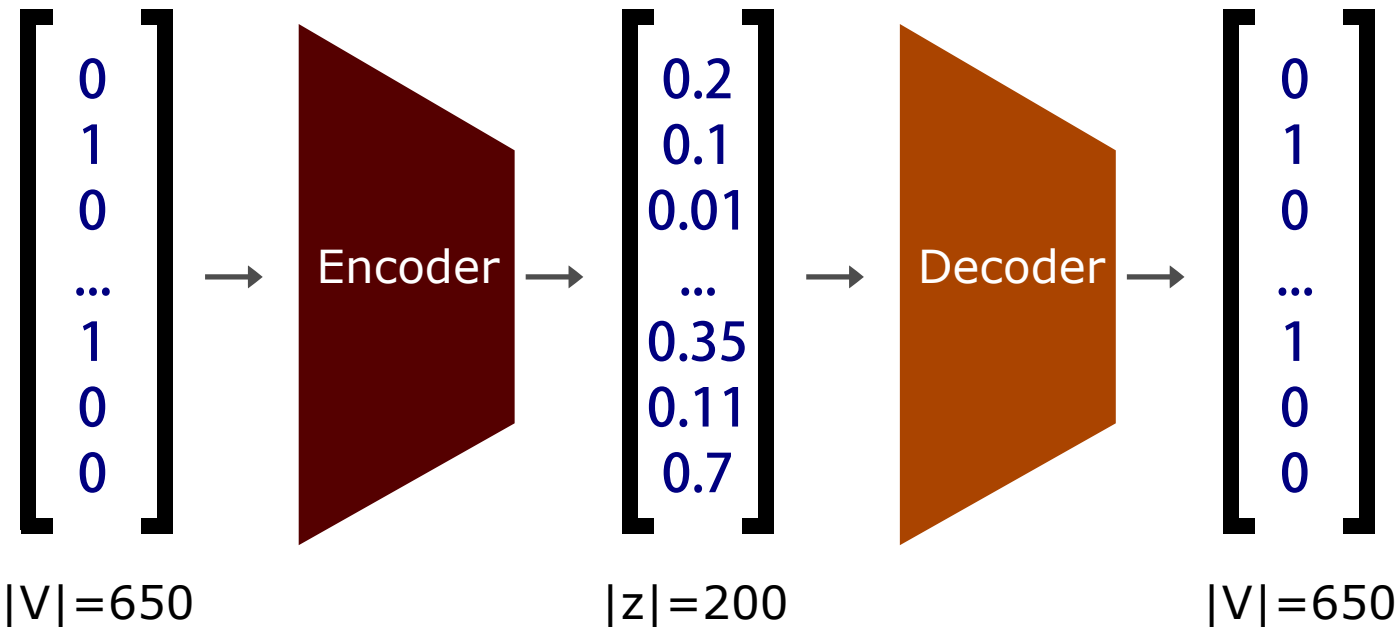
[fourier, transform, input, data, centralized, single, backend, processor, via, mc, user, interface, comply, defined, requirement, wcag, accesible, standard, ... , be, report, status]

Vocabulary using Pareto-based frequency cutoff

[fourier, transform, input, data, centralized, single, backend, processor, via, mc, user, interface, comply, defined, requirement, wcag, accesible, standard, ... , be, report, status]

Labeling using the Pareto-based vocabulary

	Sentence	Preprocessed_sentence	Bin_class(PBV)	Class
1	All Fourier Transform input data is centralized on a single Backend processor via MC defined in BE	fourier, transform, input, data, centralized, single, backend, processor, via, mc, defined, be	[0,0,1, ..., 0,1,0]	1
2	User interfaces comply with defined requirements and WCAG accessibility standards	user, interface, comply, defined, requirement, wcag, accesible, standard	[1,0,1, ..., 0,1,0]	0
	...	...	...	...
534	BE reports status, errors, and debug information to MC.	be, report, status, error, debug, information, mc	[0,0,0, ..., 0,1,0]	1
535	Navigation elements are self-descriptive and easy to understand.	navigate, element, self, descriptive, easy, understand	[0,0,0, ..., 0,0,1]	0



Loss function: Binary Cross-Entropy
Learning metric: Cosine Similarity
Activation function in Encoder: Relu
Activation function in Decoder: Sigmoid
Optimizer: Adam
Epochs: 200
Early stopping: patience =15 in val_loss

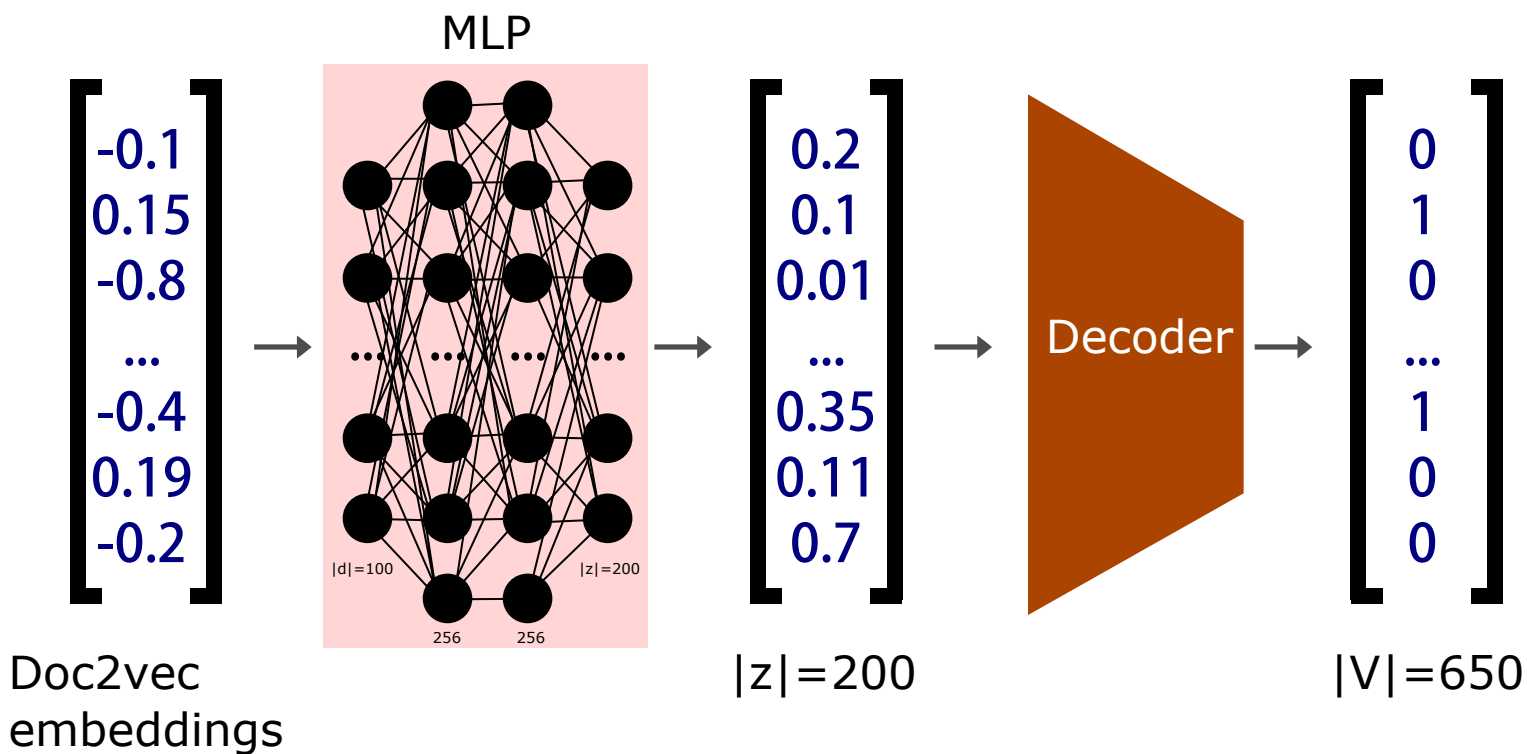
	Sentence	Preprocessed_sentence	Bin_class(PBV)	Class
1	All Fourier Transform input data is centralized on a single Backend processor via MC defined in BE	fourier, transform, input, data, centralized, single, backend, processor, via, mc, defined, be	[0,0,1, ..., 0,1,0]	1
2	User interfaces comply with defined requirements and WCAG accessibility standards	user, interface, comply, defined, requirement, wcag, accesible, standard	[1,0,1, ..., 0,1,0]	0
	...	...	...	...
534	BE reports status, errors, and debug information to MC.	be, report, status, error, debug, information, mc	[0,0,0, ..., 0,1,0]	1
535	Navigation elements are self-descriptive and easy to understand.	navigate, element, self, descriptive, easy, understand	[0,0,0, ..., 0,0,1]	0



Encode the sentence using Doc2Vec



	Sentence	Doc2Vec embedding	Bin_class(PBV)	Class
1	All Fourier Transform input data is centralized on a single Backend processor via MC defined in BE	-0.17798536 -0.15071636 ... -0.44422758 0.06981415	[0,0,1, ..., 0,1,0]	1
2	User interfaces comply with defined requirements and WCAG accessibility standards	-0.01114303 0.10467981 ... -0.19350486 0.02641434	[1,0,1, ..., 0,1,0]	0
	...	...	...	...
534	BE reports status, errors, and debug information to MC.	0.14118612 0.26249644 ... 0.08122799 0.39335814	[0,0,0, ..., 0,1,0]	1
535	Navigation elements are self-descriptive and easy to understand.	-0.16182517 -0.09649321 ... 0.10482907 0.1928615	[0,0,0, ..., 0,0,1]	0



Loss function: MSE + Cosine $\alpha = 0.7$
 $\alpha * (1.0 - \cos) + (1.0 - \alpha) * \text{mse}$
 Learning metric: Cosine Similarity
 Hidden activation functions: Relu
 Output activation function: None

Optimizer: Adam
 Epochs: 200
 Early stopping: patience =15 in val_loss